

Interview Questions — React - Fundamentals

Total Questions: 41

1. What is React and what problem does it solve in web development?

Hint: Think about how React manages UI updates compared to traditional JavaScript.

2. Why is the Virtual DOM important in React?

Hint: Focus on how it affects performance when updating the UI.

3. How do you create a new React project using create-react-app?

4. After running your first React app, how would you modify the default App component to display your name?

5. What is JSX and why is it used in React?

Hint: Consider how JSX relates to JavaScript and HTML.

6. What are the main differences between a function component and a class component in React?

Hint: Think about syntax and how state is managed.

7. Write a simple function component in React that displays 'Hello, {username}' where username is a variable passed as a prop.

8. What are props in React and how are they used to pass data between components?

Hint: Props are similar to function arguments.

9. How can you set default props for a component, and why might you want to use PropTypes?

10. Imagine you have a reusable Card component that takes a title and description as props. How would you use this component to display different cards?

11. What is state in React, and how does it differ from props?

Hint: Focus on who controls state vs props.

12. How would you use the useState hook to create a counter that increments when a button is clicked?

13. Describe the steps involved in updating state in a React component using the `useState` hook.

14. How do you handle a button click event in React?

Hint: Think about event handler functions and JSX syntax.

15. Suppose you have a form with an input field. How would you capture and display the user's input as they type?

16. What is the purpose of using `map()` when rendering lists in React?

Hint: Consider how React renders multiple elements dynamically.

17. How would you render a list of product names stored in an array using React?

18. Explain two ways to conditionally render content in React.

19. What is the `useEffect` hook used for in React?

Hint: Think about side effects and component lifecycle.

20. How can you use `useEffect` to fetch data from an API when a component mounts?

21. What is a `ref` in React, and how can you use it to focus an input field after form submission?

Hint: Refs provide direct access to DOM elements.

22. Explain the difference between controlled and uncontrolled form components in React.

23. How would you set up routing in a React app using React Router?

Hint: Think about `BrowserRouter`, `Route`, and `NavLink`.

24. What is the purpose of `NavLink` in React Router, and how does it differ from a regular `Link`?

25. Describe how you would create a multi-page navigation with Home, About, and Contact pages in React.

26. What is the `Context` API in React and when would you use it?

Hint: Consider situations where prop drilling becomes a problem.

27. How does the `useContext` hook help you access global state in a React app?

28. Describe how you would implement a dark/light theme toggle using Context API.

29. How can you make a GET request to fetch data from an API using Axios in a React component?

30. How would you send form data to a server using a POST request with Axios?

31. Suppose you want to display a list of products fetched from a mock API. What steps would you follow to fetch and render this data in React?

32. How do you handle errors when making API calls in React?

Hint: Think about using try/catch blocks and updating UI accordingly.

33. If an API call fails, how would you display an error message to the user in your React component?

34. What command do you use to create a production build of your React app, and why is this step important before deployment?

35. Explain the basic steps to deploy a React Single Page Application (SPA) on Netlify.

36. In a portfolio project, how would you design reusable components for sections like About, Projects, and Contact?

37. How could you use GitHub Copilot or ChatGPT to help you design the structure of your React components for a portfolio project, and what would you check before using their suggestions?

Hint: Think about code quality, correctness, and understanding the output.

38. Describe a situation where you might use both the useEffect and useContext hooks together in a React application.

39. How would you debug a situation where your React form is not updating the state as expected when typing in the input field?

Hint: Check if the input is controlled and if the onChange handler updates state properly.

40. Suppose you have a Card component that fetches data from an API and displays it. How would you handle loading, success, and error states in this component?

41. How could AI tools like Copilot or ChatGPT assist you when you get stuck writing a React hook or fixing a bug, and why is it important to understand and test the code they provide?

Hint: AI can suggest code, but you are responsible for verifying it works in your context.